Fisher Analysis Agent

Agentic AI for Business and FinTech | Task 1 report and Task 2 reflection

Scope and approach

The agent evaluates three Fisher points for each fictional company using LangChain and **deepseek-v4-flash**. It obtains company evidence only through the supplied tools. I selected growth, profitability and margin trend because they provide complementary, auditable views of business performance. The thresholds below are my operational rules, not numerical thresholds prescribed by Fisher.

Point 1 - Market potential for sales growth

Why: Relative growth helps distinguish company performance from an expanding industry. **PASS rule:** Cumulative FY2023-to-FY2025 revenue growth must be at least the company's own industry median two-year growth. **Evidence:** `orders`, `company` and `industry_medians.revenue_growth_2y_pct`. Historical growth is a proxy for future potential, not a forecast.

Point 5 - Worthwhile profit margin

Why: Growth is less attractive when little revenue becomes operating profit, and industry benchmarks account for sector differences. **PASS rule:** FY2025 operating margin must be at least the company's own industry median operating margin. **Evidence:** `orders`, `monthly_costs`, `company` and `industry_medians.operating_margin_pct`.

Point 6 - Maintaining or improving margins

Why: The direction of profitability adds information beyond its current level. **PASS rule:** FY2025 operating margin must be at least FY2023 operating margin; equality counts as maintenance. **Evidence:** FY2023 and FY2025 records in `orders` and `monthly_costs`. A company can pass the margin-level test while failing the trend test.

Calculations and comparability

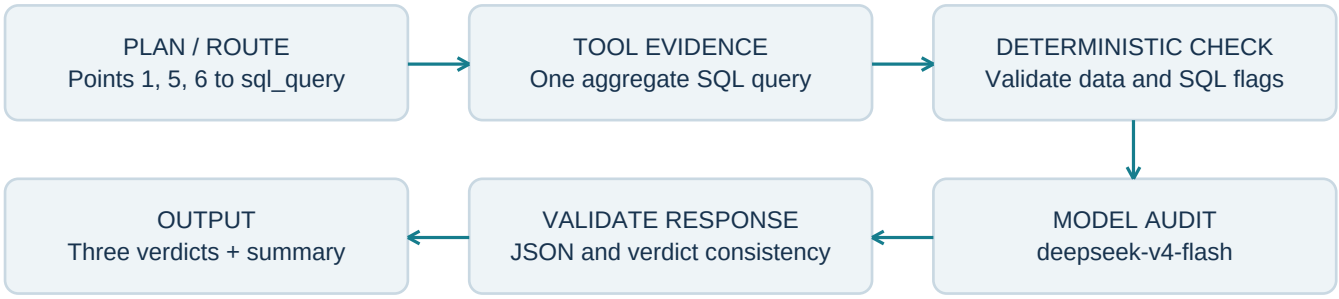
```
Revenue_y = SUM(order amounts in year y)
Costs_y = SUM(cost of sales + selling/admin + R&D in year y)
Margin_y (%) = 100 * (Revenue_y - Costs_y) / Revenue_y
Growth (%) = 100 * (Revenue_2025 / Revenue_2023 - 1)
Margin change (pp) = Margin_2025 - Margin_2023
```

All amounts use USD thousands. Orders and costs are aggregated separately before joining to avoid multiplying totals. Full calendar years are compared; partial FY2026 revenue is excluded. Ratios and comparisons are calculated in SQL before the tool rounds displayed values. Python checks positive revenue denominators, finite values and twelve months of evidence in each evaluation year.

Framework: Philip A. Fisher, *Common Stocks and Uncommon Profits*, Part One, Chapters 3-4; selected point definitions are reproduced in the course starter code.

Agent design and results

Tool-based evidence, bounded model review and reproducible public-test evaluation



Invalid model response: one repair attempt; if still invalid, retain SQL verdicts and label fallback.
Invalid evidence: return an error entry and continue. Up to four company workflows run concurrently.

How the workflow operates

A fixed plan routes all three points to one aggregate `sql_query` call per company; no interviews or management documents are needed for these numerical rules. A shared LangChain prompt-model-JSON-parser chain asks DeepSeek to review the comparisons, distinguish margin level from trend, and write a qualitative interpretation. Python accepts it only if the verdicts match the SQL flags. The numerical summary comes directly from tool evidence. One repair call is allowed; persistent model failures are labeled as fallback, while missing evidence produces no invented verdicts.

Public-test performance

Measure	Live run 1	Clean-checkout live run
Companies / correct verdicts	20 / 60 of 60	20 / 60 of 60
True-PASS / true-FAIL recall	42/42 / 18/18	42/42 / 18/18
Balanced point accuracy	100.0%	100.0%
Disagreeing companies / model fallbacks	0 / 0	0 / 0
Interviews used	0	0

Balanced accuracy = (42/42 + 18/18) / 2 = **100%**. There were **no disagreements** with the public answers. Ashgrove Labs passed growth and margin level but failed trend; Everline Dynamics passed growth and trend but failed margin level. These contrasting examples support evaluating the points separately.

Validation and limits

The second live run completed in **14.37 seconds** from a clean starter checkout with the solution applied, using the existing dependency environment and an environment-supplied key. Nine offline tests passed, covering evidence validation, rounding, failure isolation, model repair, API fallback, JSON validation, prompt/parser integration, routing and missing keys. The supplied tools, runner and scorer remain unchanged; the agent contains no company-specific answers.

Public accuracy does not guarantee private-test accuracy or investment performance. Rules were checked against public examples, so this is not an independent out-of-sample evaluation. Margin endpoints miss intervening volatility and do not explain management actions. Valid JSON and matching verdicts also do not prove that every interpretive sentence is justified.

Reproducibility: run `python3 hw3.py --data-folder public_test`. Evidence: `validation/public_test_results.csv`, `validation/clean_public_test_results.csv` and `validation/clean_run.txt`.

TASK 2 / 03

Reflection

AI sycophancy, reinforcing conversations and evidence-grounded analysis

Agreement should not replace evidence

An AI assistant can be fluent, friendly and wrong at the same time. I understand AI sycophancy as a tendency to agree with a user's beliefs or preferences even when evidence does not support them. In business analysis, a system that makes an investor feel confident may be less useful than one that identifies a weak assumption. Analytical assistance should improve a decision, including when that requires disagreement.

Suppose a user asks, "This company is clearly excellent; explain why I am right." An agreeable answer could highlight revenue growth while ignoring deteriorating margins. Every highlighted fact might be correct, yet the selection of facts could mislead. The problem therefore includes selective attention and unjustified certainty, as well as factual hallucination.

The Fisher agent resists part of this pressure by fixing rules before evaluation and separating margin level from margin trend. A company can have an above-industry margin that is declining. Preserving this mixed conclusion is more useful than forcing every verdict into a reassuring story. Model verdicts are checked against SQL comparisons, and inconsistent responses are rejected.

When a conversation becomes a feedback loop

I interpret the assignment's term "AI-psychosis" cautiously, as a prompt to consider conversations that reinforce unfounded beliefs or weaken connection to outside evidence. I would not use a transcript or this term to diagnose someone. My concern is interaction design: speculation may repeatedly be presented as confirmation, while confident language is mistaken for independent verification.

For example, a user might believe ordinary market movements contain secret messages intended specifically for them. An assistant that invents a detailed explanation could turn an unsupported interpretation into an apparently authoritative narrative. Repeated agreement may make the idea feel better supported without introducing new evidence. A caring response should acknowledge feelings without confirming the unsupported explanation, express uncertainty, and leave room for outside perspectives and human support.

Sycophancy and such reinforcing interactions are related but not interchangeable. Sycophancy also affects ordinary investment memos without a mental-health context. Conversational success should not be measured only by whether a user likes the answer. Useful assistance can mean slowing down, asking what would disprove a claim, or declining to invent an explanation.

What this changes in my agent design

I would test the agent with neutral, strongly optimistic and strongly pessimistic prompts while keeping company evidence identical. Verdicts should remain stable. I would also test whether it states "insufficient evidence," identifies conflicting signals and distinguishes observations from forecasts. These are proposed behavioral tests, not experiments already completed here.

The implementation prevents the model from changing numerical rules and reports failures explicitly. It cannot make a poorly chosen rule valid, nor guarantee that a plausible explanation is justified. Public-test accuracy therefore does not establish investment validity. Human review remains necessary for assumptions, evidence and interpretation. The standard I would apply is whether the assistant helps a user think more carefully, rather than merely feel more certain.

Preparation note: This report and reflection were prepared with AI assistance. Examples in the reflection are hypothetical, not personal experiences or clinical cases.